

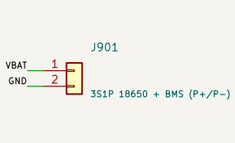
A3 Lite module wiring for a perfboard build; NOT a PCB. Verify each breakout pin order before soldering.

E-stop (NC) cuts both TB6612 VM and both servo buck inputs; ESP32, eyes and ToF stay powered and report.

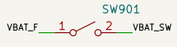
ACT_EN (10k pulldown) = TB6612 STBY x2 + both servo buck EN; PCA9685 /OE pulled up: actuators off in reset.

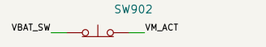
E-stop sense = its own 2nd NC contact (fail-safe High), edge-latched; 7.5R ballast bounds motor stall < 1.0 A.

Motor speed: MCPWM on IN1/IN2 with PWMA/PWMB high (drive/coast). Servo /OE High before ACT_EN Low.

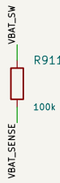






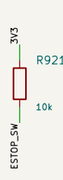






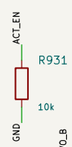








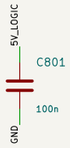


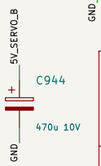


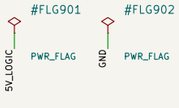












A3 Lite wiring P0 (M3): breakout pin orders, Mini560 EN level and PCA9685 header to verify

Sheet: /
File: lite_wiring.kicad_sch

Title: UGV-01 A3 Lite module wiring

Size: A3	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 1/1